2022 RI H2 Chemistry Prelim Paper 3 – Suggested Solutions

Section A

1(a)(i)

$$[CO_3^{2-}] = \frac{0.010 \times \frac{20}{1000}}{\frac{20+20}{1000}} = 5.00 \times 10^{-3} \text{ mol dm}^{-3}$$

As maximum amount of AgCl have been precipitated without precipitating Ag_2CO_3, the solution is saturated with Ag_2CO_3. Hence,

ionic product = $K_{sp}(Ag_2CO_3)$
$[Ag^+]^2(5.00 \times 10^{-3}) = 8.1 \times 10^{-12}$
$[Ag^+] = 4.025 \times 10^{-5}$ mol dm^{-3}

As the solution is also saturated with AgCl,
$[Ag^+][Cl^-]_{remaining} = (4.025 \times 10^{-5})[Cl^-]_{remaining} = 2.0 \times 10^{-10}$
$[Cl^-]_{remaining} = \underline{4.97 \times 10^{-6} \text{ mol dm}^{-3}}$

1(a)(ii)

$$\text{Initial } [Cl^-] \text{ in mixture} = \frac{0.010 \times \frac{20}{1000}}{\frac{20+20}{1000}} = 5.00 \times 10^{-3} \text{ mol dm}^{-3}$$

Percentage of Cl^- precipitated

$$= \frac{(5.00 \times 10^{-3}) - (4.97 \times 10^{-6})}{5.00 \times 10^{-3}} \times 100\%$$

= <u>99.9%</u>

This is <u>an effective</u> method.

1(b)(i) Co-ordination number is the <u>number of</u> nearest neighbouring <u>ions that surrounds an ion of opposite charge</u>.

1(b)(ii) As the <u>Ba^{2+} ion is larger</u> than the Ca^{2+} ion, the Ba^{2+} ion is <u>able to accommodate a greater number of chloride ions surrounding it</u>.

1(b)(iii) The ions in $CaCl_2$ are <u>less tightly packed</u> in the lattice structure. Hence, <u>less energy</u> is required to overcome the weaker electrostatic forces of attraction between Ca^{2+} and Cl^- ions.

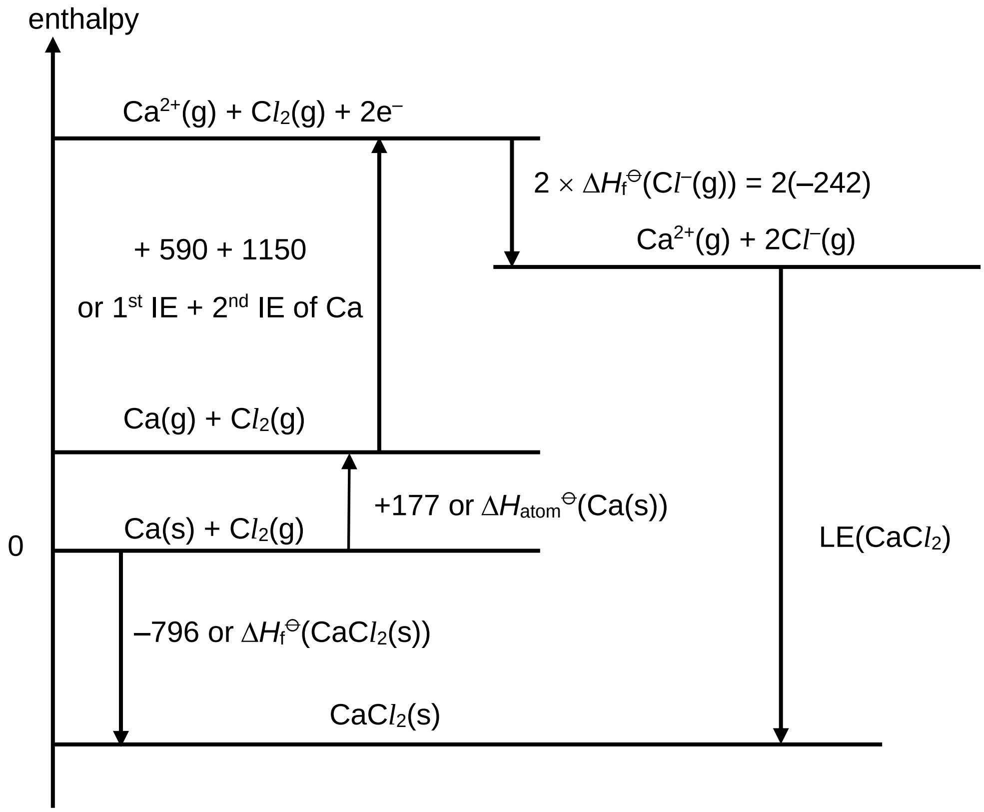
OR

Due to its smaller size/radius, Ca^{2+} has a higher charge density and polarising power than Ba^{2+}. As such, $CaCl_2$ has a <u>greater covalent character</u> than $BaCl_2$, resulting in an unexpectedly lower melting point than that of $BaCl_2$.

1(c)(i) <u>As the process occurs readily (or is spontaneous)</u>, <u>ΔG is negative</u>. Since there is a <u>decrease in the number of ways the water particles can be arranged</u>, <u>ΔS is negative</u>. $\Delta G = \Delta H - T\Delta S$. Since $-T\Delta S$ is positive, ΔH_1 must be <u>negative</u>.

1(c)(ii) The lattice energy (LE) of calcium chloride is the energy released when 1 mole of solid $CaCl_2$ is formed from $Ca^{2+}(g)$ and $Cl^-(g)$ at 1 bar and 298 K.

1(c)(iii)



$+177 + 590 + 1150 + 2(-242) + \text{LE}(CaCl_2) = -796$
$\text{LE}(CaCl_2) = \underline{-2230 \text{ kJ mol}^{-1}}$

1(d) $AlCl_3$ exists as simple covalent molecules but dissolves in water to form $[Al(H_2O)_6]^{3+}$ ions.

$AlCl_3(s) + 6H_2O(l) \longrightarrow [Al(H_2O)_6]^{3+}(aq) + 3Cl^-(aq)$

In $[Al(H_2O)_6]^{3+}$, the high charge density of Al^{3+} polarises and weakens the O–H bond of the water molecules, allowing the complex ion to undergo hydrolysis to form H_3O^+. A weakly acidic solution of pH 3 results.

$[Al(H_2O)_6]^{3+}(aq) + H_2O(l) \rightleftharpoons [Al(H_2O)_5(OH)]^{2+}(aq) + H_3O^+(aq)$

PCl_5 exists as simple covalent molecules, which hydrolyses in water as the low-lying d-orbitals of the central P atom accepts a lone pair of electrons from H_2O. The reaction produces HCl as a product which dissolves in water to form a strongly acidic solution of pH 2.

$PCl_5(s) + 4H_2O(l) \longrightarrow H_3PO_4(aq) + 5HCl(aq)$

2(a)(i) Cyclopentadienyl anion consists of a ring of five sp^2 hybridised carbon atoms. Each carbon atom has an unhybridised p orbital that overlaps continuously.

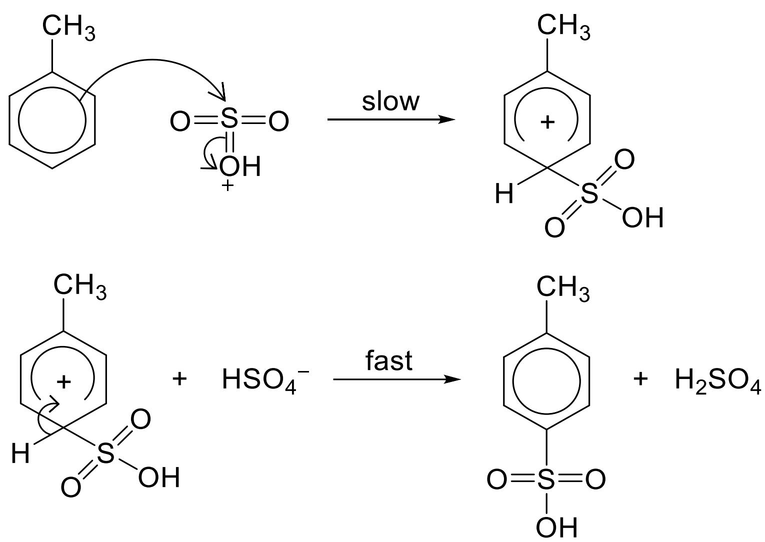
The C atom bearing the negative charge contains a lone pair of electrons in its unhybridised p orbital. Hence, together with the 4 π electrons from the two C=C bonds, the cyclopentadienyl anion has a total of 6 delocalised π electrons.

2(a)(ii) When naphthalene undergoes substitution reaction, the product formed retains its aromaticity (of the entire molecule) as there are 10 delocalised π electrons in the continuously overlapping p orbitals.

However, if naphthalene undergoes addition reaction, the product formed is no longer aromatic (across the entire molecule) due to the absence of continuously overlapping p orbitals (or absence of 4n + 2 delocalised π electrons).

2(b)(i) $SO_3 + H_2SO_4 \rightleftharpoons HSO_3^+ + HSO_4^-$

2(b)(ii) Electrophilic substitution

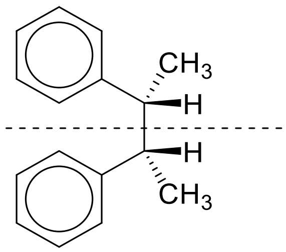


2(b)(iii) concentrated H_2SO_4, concentrated HNO_3, heat (under reflux)

2(c) Nitrogen is less electronegative than oxygen and its lone pair of electrons is more readily delocalised into the benzene ring. Hence, -NH_2 group is more activating than the -OH group and directs the electrophilic substitution to occur more at the 2-position relative to -NH_2.

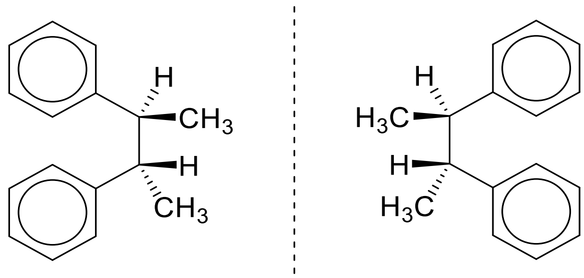
2(d)(i) (excess) benzene, $FeBr_3$ (or $AlBr_3$)

2(d)(ii)

Structure of **X**: 

plane of symmetry

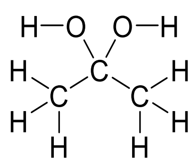
As **X** has a <u>plane of symmetry</u> / <u>is a meso compound</u>, **X** is optically inactive.

Structures of **Y** and **Z**: 

mirror plane

Y and **Z** are <u>a pair of enantiomers</u> and hence rotates plane-polarised light in equal but opposite directions.

2(e)(i)



2(e)(ii) Let the molecular formula of **K** be $C_xH_yO_z$.

$$C_xH_yO_z + (x + \frac{y}{4} - \frac{z}{2})O_2 \longrightarrow xCO_2 + \frac{y}{2} H_2O$$

n(H_2O) collected = $\frac{0.090}{18.0}$ = <u>0.005 mol</u>

Vol of gases (after cooling) = V(unreacted O_2) + V(CO_2) = 202 cm^3

V(CO_2) = 144 cm^3
n(CO_2) evolved = $\frac{144}{24000}$ = <u>0.006 mol</u>

V(unreacted O_2) = 202 − 144 = 58 cm^3
V(reacted O_2) = 250 − 58 = 192 cm^3
n(reacted O_2) = $\frac{192}{24000}$ = <u>0.008 mol</u>

	$C_xH_yO_z$	O_2	CO_2	H_2O
amount / mol	0.001	0.008	0.006	0.005
mole ratio	1	8	6	5

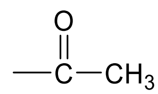
$x = 6$

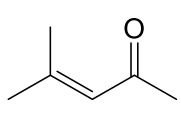
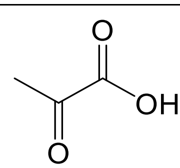
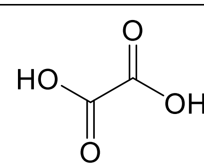
$\frac{y}{2} = 5$, hence $y = 10$

$x + \frac{y}{4} - \frac{z}{2} = 6 + 2.5 - \frac{z}{2} = 8$, hence $z = 1$

Molecular formula for **K** is $C_6H_{10}O$.

2(e)(iii)

Evidence / Information	**Deduction / Explanation**
K gives an orange ppt with 2,4-DNPH but does not form silver mirror with Tollens' reagent.	Condensation with 2,4-DNPH but no oxidation with Tollens' reagent. **K** is a carbonyl compound but not an aldehyde. **K** is a ketone.
When **K** is heated with acidified $KMnO_4$, **L** and propanone are formed.	Strong oxidation **K** is an alkene that contains $=C(CH_3)_2$ group.
When **L** is heated with aqueous I_2 in dilute NaOH followed by acidification, **M** and a yellow ppt are formed.	Oxidation and acid-base reaction **L** has 3 C atoms and contains  group. Yellow ppt is CHI_3. **M** contains a –COOH group.
When **M** is heated with acidified $KMnO_4$, the only product form is a gas that gives a white precipitate with limewater.	Oxidation Gas is CO_2.

		
K	**L**	**M**

3(a)(i) From the *Data Booklet*, $E^\ominus_{cathode}$ = +0.40 V and $E^\ominus_{anode}$ = 0.00 V.
Hence, $E^\ominus_{cell} = E^\ominus_{cathode} - E^\ominus_{anode}$ = +0.40 V

3(a)(ii) $O_2 + 2H_2O + 4e^- \rightleftharpoons 4OH^-$ (a)
$2H^+ + 2e^- \rightleftharpoons H_2$ (b)

At the cathode, CO_2 removes OH^- formed, causing a decrease in $[OH^-]$. The position of equilibrium for reaction (a) shifts to the right and this causes $E_{cathode}$ to be more positive.

At the anode, the CO_3^{2-} which migrates from the cathode removes H^+ formed, causing a decrease in $[H^+]$. The position of equilibrium for reaction (b) shifts to the left and this causes E_{anode} to be more negative.

Hence, E_{cell} of the EDCS is more positive than that of the hydrogen-fuel cell in **(a)**.

3(a)(iii) The higher the CO_2 concentration, the more positive the E_{cell} and the current increases. Since $I = n_eF/t$, the amount of electrons transferred per second increases and hence increases the rate of removal of CO_2.

3(b)(i) From the *Data Booklet*,
If a typical hydrogen-oxygen fuel cell operates under acidic conditions:
cathode reaction: $O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l)$
anode reaction: $H_2(g) \longrightarrow 2H^+(aq) + 2e^-$

$E^\ominus_{cathode}$ = +1.23 V and $E^\ominus_{anode}$ = 0.00 V
Hence, $E^\ominus_{cell}$ = +1.23 − 0.00 = +1.23 V

If a typical hydrogen-oxygen fuel cell operates under alkaline conditions:
cathode reaction: $O_2(g) + 2H_2O(l) + 4e^- \longrightarrow 4OH^-(aq)$
anode reaction: $H_2(g) + 2OH^-(aq) \longrightarrow 2H_2O(l) + 2e^-$

$E^\ominus_{cathode}$ = +0.40 V and $E^\ominus_{anode}$ = −0.83 V
Hence, $E^\ominus_{cell}$ = +0.40 − (−0.83) = +1.23 V

3(b)(ii) Under acidic conditions, there will not be any OH^- ions at the cathode to react with and to convert atmospheric CO_2 to CO_3^{2-}. Hence, the CO_2 in air cannot be captured by the EDCS.

Under alkaline conditions, there will not be any H^+ ions at the anode to convert CO_3^{2-} ions back to CO_2 to be removed into a separate storage.

Hence, a typical hydrogen-oxygen fuel cell cannot be used in the EDCS.

3(c)(i)

Volume of CO_2 removed per minute = $0.99 \times 3000 \times \dfrac{400}{10^6} = 1.188 \text{ cm}^3$

Amount of CO_2 removed per second = $\dfrac{1.188}{24000} \div 60 = 8.25 \times 10^{-7}$ mol

From equations (1) – (4),
amount of electrons passed per second
$= 2 \times 8.25 \times 10^{-7}$
$= 1.65 \times 10^{-6}$ mol per second

Hence, current = $1.65 \times 10^{-6} \times 96500$ = $\underline{0.159 \text{ A}}$

3(c)(ii) The CO_3^{2-} and electrons can diffuse more quickly across a <u>larger surface area</u> and hence <u>increase the rate of CO_2 removal</u>.

3(d) Similar to graphite, CNTs have an <u>**extended** delocalised π electron cloud</u>, which allows CNTs to conduct electricity.

Thus, it is likely that CNTs help conduct electricity across the membrane and <u>increases the membrane's electrical conductivity</u>.

3(e)(i) Mg^{2+} has a smaller ionic radius than Ca^{2+}. Hence, with the same charge, Mg^{2+} has a <u>higher charge density and hence greater polarizing power</u> than Ca^{2+}. As a result, <u>Mg^{2+} distorts the electron cloud of R-COO$^-$ to a greater extent</u>. The <u>covalent bonds within R-COO$^-$</u> are <u>weakened to a greater extent</u> as compared to that in $(R\text{-}COO^-)_2Ca^{2+}$. Therefore, <u>less heat energy</u> is needed to decompose $(R\text{-}COO^-)_2Mg^{2+}$. Hence $(R\text{-}COO^-)_2Mg^{2+}$ has <u>lower thermal stability</u>.

3(e)(ii) $MgCO_3 \longrightarrow MgO + CO_2$

3(e)(iii) $(CH_3\text{-}COO^-)_2Mg^{2+} \longrightarrow CH_3COCH_3 + CO_2 + MgO$

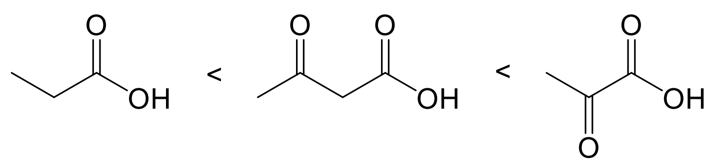
Section B

4(a) Nitrogen has a higher number of protons and hence higher nuclear charge than carbon. Although there is one more electron in nitrogen than carbon, this electron is added to the same outermost shell, and hence shielding effect remains approximately constant. Thus, nitrogen has a higher effective nuclear charge and stronger electrostatic attraction between the nucleus and the valence electrons than carbon, resulting in an increase in the amount of energy required to remove the valence electron from nitrogen. Hence, the first ionisation energy of nitrogen is higher than carbon.

The first ionisation energy of oxygen is lower than that of nitrogen. This is because the 2p electron to be removed from oxygen is a paired electron while that to be removed from nitrogen is an unpaired electron. Due to inter–electronic repulsion between paired electrons in the same orbital, less energy is required to remove the paired 2p electron from oxygen.

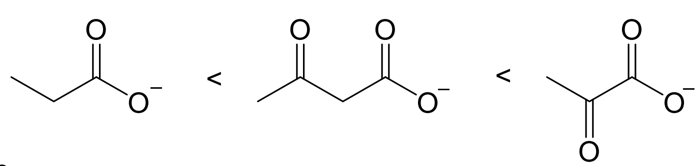
4(b) The stronger the acid, the lower the pK_a value and the more stable the conjugate base.

Based on the given pK_a values, acidity increases in the order:



F **D** **E**

and the stability of the conjugate base increases in the order:

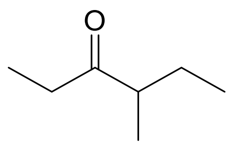


conjugate base of: **F** **D** **E**

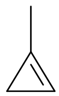
Compared to the conjugate base of **F**, the conjugate bases of **D** and **E** are more stable due to the presence of the electron-withdrawing carbonyl group to disperse the negative charge and hence stabilising the anions.

The conjugate base of **E** is more stable than that of **D** as the electron-withdrawing carbonyl group is nearer to the $–COO^–$, and can better disperse the negative charge, thus stabilising the conjugate base of **E** to a greater extent.

4(c)(i)

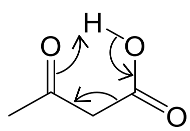


4(c)(ii) step 1: ethanolic KOH, heat (under reflux)

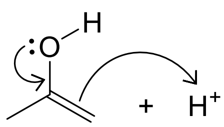
I : 

step 2: $KMnO_4$(aq), H_2SO_4(aq), heat (under reflux)

4(c)(iii) step 1:



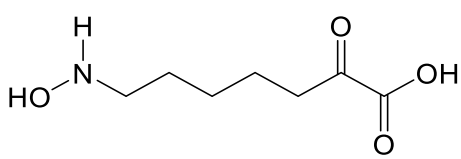
step 2:



4(d)(i) Acidic hydrolysis and oxidation

4(d)(ii) $KMnO_4$(aq), H_2SO_4(aq), heat OR $K_2Cr_2O_7$(aq), H_2SO_4(aq), heat

4(e)(i)

J: 

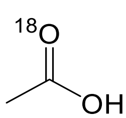
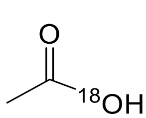
4(e)(ii) High dilution or low concentration of reactant

4(f)(i) A catalytic amount of H^+ is not sufficient as H^+ is <u>consumed</u> stoichiometrically <u>in step 6</u> to protonate the NH_3 hydrolysis product.

4(f)(ii) In conjugate acid **X**, the <u>lone pair of electrons</u> on the N atom <u>interacts with the π electron cloud of the adjacent C=O bond</u> and is <u>delocalised</u>, hence <u>dispersing the positive charge on the O atom</u> and stabilising **X**.

In conjugate acid **Y**, there is no continuous side-on overlap of p orbitals and hence **Y** is <u>not resonance stabilised</u>.

4(f)(iii)

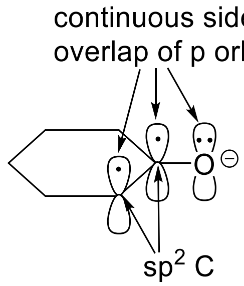
 and 

5(a)(i) Cyclohexanone and compound **P** are <u>constitutional (or structural) isomers</u>.

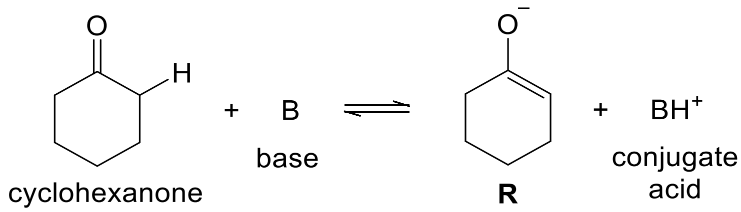
5(a)(ii) Compound **Q** is more stable than pentane-2,4-dione because:

- the H atom of the <u>–OH group</u> can form <u>intramolecular hydrogen bond</u> with the O atom of the nearby <u>carbonyl group</u>.
- there is <u>resonance stabilisation</u> / <u>delocalisation of lone pair of electrons on the O atom of -OH group with the π electron cloud of the C=C and C=O</u> groups.

5(b)(i) The carbon atoms in the C=C bond are sp^2 hybridised, each with an unhybridised p orbital containing one electron. The <u>p orbital</u> of the adjacent <u>O atom can overlap continuously side-on with the p orbitals of the sp^2 carbon atoms</u>, resulting in <u>delocalisation of the negative charge on the O atom into the C=C bond</u>. Hence, **R** is resonance stabilised.



5(b)(ii)



Unlike conjugate acids of **I** and **II**, the pK_a of the conjugate acids of **III** and **IV** are higher than the pK_a of cyclohexanone. This means that the conjugate acids of **III** and **IV** are weaker acids than cyclohexanone. Hence, the equilibrium position of the acid-base reaction of **III** and **IV** with cyclohexanone lie further right compared to that of **I** and **II** with cyclohexanone.

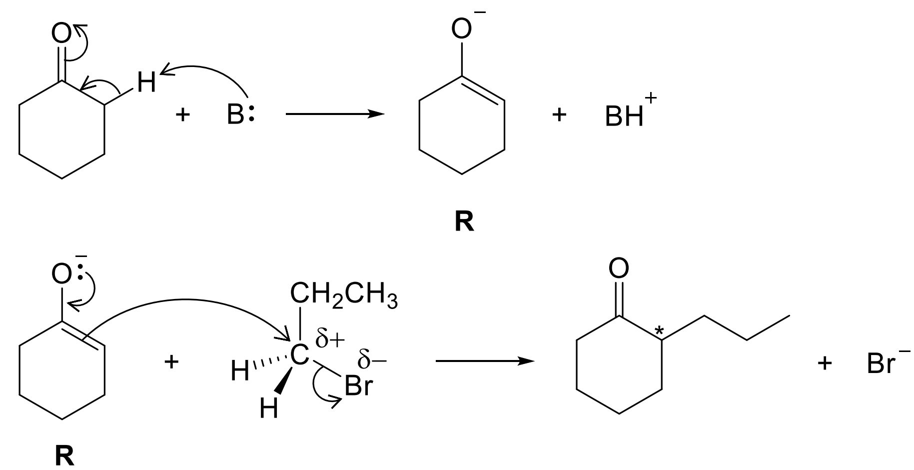
Compared to **IV**, **III** is more suitable as the <u>presence of bulky alkyl groups in **III** prevents it from acting as a nucleophile</u> to react with the carbonyl group in cyclohexanone, resulting in the formation of side/addition product.

Hence, **III** is the most suitable base to deprotonate cyclohexanone to form **R**.

5(c)(i)

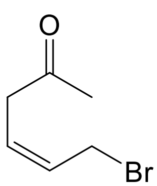
step 1

step 2



5(c)(ii) In step 2, the S_N2 reaction can occur with 1-bromopropane on <u>either side of the trigonal planar carbon</u> of C=C in **R** with <u>equal probability</u>, hence forming a <u>racemic mixture / equal amounts of a pair of enantiomers</u> which does not rotate the plane of polarised light.

5(c)(iii)



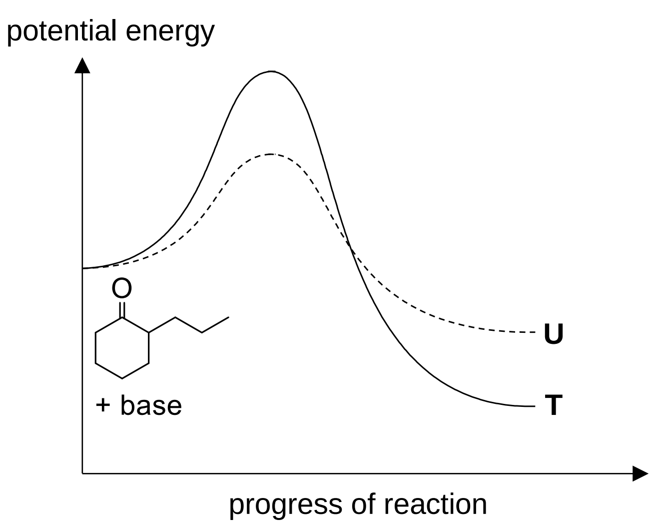
S

C_6H_9BrO

5(c)(iv) step 2: H_2, Ni, heat

step 3: $KMnO_4$(aq), H_2SO_4(aq), heat (under reflux)
OR
$K_2Cr_2O_7$(aq), H_2SO_4(aq), heat (under reflux)

5(d)



5(e)(i) Step 1 is a condensation reaction.

5(e)(ii) Compound **V** does not exhibit cis-trans isomerism. Although there is restricted rotation about the C=N bond, the C atom in the C=N bond has two identical groups bonded to it.

5(e)(iii) H_2SO_4(aq) / HC*l*(aq) / HNO_3(aq), heat
OR
NaOH(aq), heat